Abdulrahman Diaa

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Awards

2026-2027: NSERC Canada Graduate Research Scholarship - Doctoral - CA$120K *University of Waterloo*
2026-2027: Interop Labs PhD Fellowship - CA$110K *University of Waterloo*
2026-2027: President's Graduate Scholarship - CA$10K *University of Waterloo*
2021-2025: David R. Cheriton Graduate Scholarship - CA$40K *University of Waterloo*
2021-2023: International Masters Award of Excellence - CA$12.5K *University of Waterloo*
2021-2022: Johnston International Entrance Scholarship - CA$5K *University of Waterloo*
2016-2021: Tarek Nour AUC Scholarship - US$100K *The American University in Cairo*

Research Experience

Student Researcher **Ontario, Canada**
Google *Apr 2026 - Dec 2026*
Topics: Private RAG and Nearest Neighbor Search
Reference: Sarvar Patel

Cybersecurity Researcher **München, Germany**
Airbus Defence and Space – VCX *Jan 2025 - July 2025*
Topics: Practical Batch Private Information Retrieval and Privacy-preserving DBSCAN
Reference: Dr. Erik-Oliver Blass

Research Intern **Lausanne, Switzerland**
École Polytechnique Fédérale de Lausanne – Lab for Data Security (LDS) *July 2021 - Sept 2021*
Topic: Privacy-preserving Federated Learning for Recurrent Neural Networks
Reference: Prof. Jean-Pierre Hubaux

Undergraduate Data Science Research Intern **Cairo, Egypt**
Dell Technologies – Data Office Team *Feb 2020 - June 2021*
Topics: Technology Forecasting and Distributed Neural Networks
Reference: Eng. Steve Todd

Undergraduate Researcher **Cairo, Egypt**
The American University in Cairo *Sept 2019 - Jan 2020*
Topic: Deep Reinforcement Learning for Traffic Control
Reference: Dr. Cherif Salama

Research Software Development Engineering Intern **Cairo, Egypt**
Microsoft Advanced Technology Lab in Cairo – Document Knowledge Extraction Team *Jul 2019 - Sept 2019*
Topic: Unsupervised Clustering for Schema-Inference from Semi-Structured Documents
Reference: Eng. Achraf Chalabi

Publications

[1] **A. Diaa**, T. Aremu, and N. Lukas. "Optimizing Adaptive Attacks against Watermarks for Language Models". In: The 42nd International Conference on Machine Learning. 2025.
- Spotlight paper at ICML'25 (Top 2.6%)
- Oral presentation at The 1st Workshop on GenAI Watermarking at ICLR'25 (Top 3.9%)
- Best poster award at Cybersecurity and Privacy Institute Conference CPI'24.

[2] **A. Diaa**, T. Humphries, and F. Kerschbaum. "FastLloyd: Federated, Accurate, Secure, and Tunable k-Means Clustering with Differential Privacy". In: The 34th USENIX Security Symposium. 2025.

[3] **A. Diaa** et al. "Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions". In: The 33rd USENIX Security Symposium. 2024.

[4] N. Lukas, **A. Diaa**, L. Fenaux, and F. Kerschbaum. "Leveraging Optimization for Adaptive Attacks on Image Watermarks". In: The 12th International Conference on Learning Representations. 2024.

[5] R. Mahdavi, **A. Diaa**, and F. Kerschbaum. "ZipPIR: High-throughput Single-server PIR without Client-side Storage". In: The 35th USENIX Security Symposium. 2026.

[6] S. Sav, **A. Diaa**, A. Pyrgelis, JP. Bossuat, and JP. Hubaux. "Privacy-Preserving Federated Recurrent Neural Networks". In: The 23rd Privacy Enhancing Technologies Symposium. 2023.

Patents

Dell Technologies – Patent No: US 11934487B2 (Granted)
Splitting Neural Networks on Multiple Edge Devices to Train on Vertically Distributed Data — *Mar 2024*

Dell Technologies – Application No: US 17/314315 (Pending)
Data-Driven Index for Identification and Ranking of Companies for a Selected Technology — *May 2021*

Dell Technologies – Application No: US 17/160782 (Pending)
Forecasting Technology Phases Using Unsupervised Clustering with Wardley Maps — *Jan 2021*

Education

University of Waterloo – CrySP Lab — **Ontario, Canada**
Sept 2021 - Aug 2027

PhD in Computer Science (Sept 2023 - Aug 2027) – **GPA: 95.5%**
Thesis: *On the Traceability of Generative Models*
MMath in Computer Science (Sept 2021 - Aug 2023) – **GPA: 96.75%**
Thesis: *Differentially-private multiparty Clustering*
Supervisor: Prof. Florian Kerschbaum

The American University in Cairo — **Cairo, Egypt**
Sept 2016 - June 2021

Bsc in Mathematics (Sept 2017 - June 2021) – **GPA: 4.0/4.0**
Thesis: *Lattice-based Cryptography and Fully Homomorphic Encryption: A Survey*
Bsc in Computer Engineering (Sept 2016 - June 2021) – **GPA: 4.0/4.0**
Thesis: *Mixture of Experts for Human Activity Classification from Images*
Supervisors: Dr. Ahmed El-Guindy, Dr. Cherif Salama and Dr. Mohammed Moustafa

Academic Service

2026: Reviewer — *40th Annual Conference on Neural Information Processing Systems (NeurIPS)*
2025: Workshop Reviewer — *1st Workshop on GenAI Watermarking (WMark@ICLR)*
2025: Artifact Reviewer (Ninja Award) — *34th USENIX Security Symposium*
2024: Journal Reviewer — *Transactions on Knowledge and Data Engineering (TKDE)*

Extra-Curricular Activities

2017: Academic Vice-president — *Robotics Club, The American University in Cairo*
2017: ROV Control Engineer — *Robotics Club, The American University in Cairo*
2017: Robotics Instructor — *Robotics Club, The American University in Cairo*
2017: Content Editor — *Geniuses (TV Show)*
2016: Helped raise ∼**4M USD** for public schools — *Geniuses (TV Show)*
2016: Geniuses (TV Show) Team Leader — *Al-Bashaer International Schools*

Teaching Experience

Teacher Assistant — **Ontario, Canada**
Uinversity of Waterloo — *Jan 2023 - Apr 2023*
Course: Privacy, Cryptography, Network, and Data Security
Reference: Thomas Humphries

Teacher Assistant — **Cairo, Egypt**
The American University in Cairo — *Jan 2021 - June 2021*
Course: Embedded Systems Laboratory
Reference: Dr. Suzanne Safwat

Teacher Assistant — **Cairo, Egypt**
The American University in Cairo — *Sept 2020 - June 2021*
Course: Linear Algebra
Reference: Dr. Ahmed El-Guindy

Teacher Assistant — **Cairo, Egypt**
The American University in Cairo — *Sept 2017 - Jun 2019*
Course: Programming Fundamentals
Reference: Dr. Howaida Ismail

Other Honors

2021: President Cup — *The American University in Cairo*
2016-2021: Dean's list of Academic Standing — *The American University in Cairo*
2020: Challenge Coin for Innovation — *Dell Technologies*
2019-2020: Finalist — *Egyptian Collegiate Programming Contest*
2016: Best Player Cup — *Geniuses (TV-Show), Season 1*

Engineering Experience

Software Engineering Intern — **Cairo, Egypt**
Nokia EG — *Aug 2018 - Sept 2018*

Software Engineering Intern — **Cairo, Egypt**
Microsoft EG — *Jul 2017 - Aug 2017*